III Semester			
S. No	**Course Code**	**Course Title**	**Credit Hours**
1.	**GPB-211**	Seed Production Technology*	2(0+2)
2.	**GPB-212**	Principle of Genetics	3(2+1)
3.	**AGEXT-211**	Fundamentals of Extension Education	2(1+1)
4.	**AGRON-211**	Crop Production Technology–I (*Kharif* crops)	3(1+2)
5.	**AGRON-212**	Principles and Practices of Natural Farming	2(1+1)
6.	**AGINFO-211**	Agriculture Informatics and Artificial Intelligence	3(2+1)
7.	**HORT-211**	Production Technology for Fruit and Plantation Crops	2(1+1)
8.	**PPATH-211**	Fundamentals of Nematology	2(1+1)
9.	**PHED-211**	Physical Education, First Aid and Yoga Practices	2(0+2)
10.	**NCC-III/ NSS-III /PHED-III**	NSS/NCC/Physical Education &Yoga Practices**	1(0+1)
		TOTAL	**21(9+12)+1****
***Skill Enhancement Courses;**NC: Non-gradial courses**			

GPB-211	**Seed Production Technology***	**2(0+2)**

Practical

Planning of Seed Production, requirements for different classes of seeds in field crops - unit area and rate

Operation and handling of mechanical drying equipment; effect of drying temperature and duration on seed germination and storability seed processing equipment; seed treating equipment.

Seed production in cross pollinated crops with special reference to land, isolation, Planting ratio of male and female lines, synchronization of parental lines and methods to achieve synchrony; supplementary pollination, pollen storage, hand emasculation and pollination in tomato, Hybrid seed production in Maize, detasseling in maize, identification of rogues and pollen shedders, Pollen collection, storage, viability and stigma receptivity; gametocide application and visits to seed production plots etc., Visit to seed processing plant and commercial controlled and uncontrolled Seed Stores, Seed industries and local entrepreneurships visit to nearby areas, Different methods of examination of seeds to assess seed-borne microorganisms and to quantify infection percentage, detection of seed-borne fungi, bacteria and viruses, identification of storage fungi, control of seed- borne diseases, seed treatment methods., Maintenance of aseptic conditions and sterilization techniques, Preparation of nutrient stocks for synthetic media, Selection of explants for callus induction, Preparation of MS medium for micro-propagation and Callus induction, Selection of explants for callus induction, Preparation of MS medium for micro-propagation and Callus induction, Inoculation of explants for micro-propagation, Inoculation of explants for callus induction and subsequently regeneration of plantlets from matured seeds of field and horticultural crops, Synthetic seed preparation.

Lecture Schedule: Practical

S.N.	Topic	No. of lectures
1	Planning of Seed Production, requirements for different classes of seeds in field crops - unit area and rate	1
2	Operation and handling of mechanical drying equipment; effect of drying temperature and duration on seed germination and storability seed processing equipment; seed treating equipment.	1
3	Seed production in cross pollinated crops with special reference to land, isolation, Planting ratio of male and female lines, synchronization of parental lines and methods to achieve synchrony	1
4	supplementary pollination, pollen storage, hand emasculation and pollination in tomato	1
5	Hybrid seed production in Maize, detasseling in maize	1
6	identification of rogues and pollen shedders, Pollen collection, storage, viability and stigma receptivity; gametocide application and visits to seed production plots etc.	1
7	Visit to seed processing plant and commercial controlled and uncontrolled Seed Stores	1
8	Seed industries and local entrepreneurships visit to nearby areas	1
9	Different methodsof examination of seeds to assess seed-borne microorganisms and to quantify infection percentage	1
10	Detection of seed-borne fungi, bacteria and viruses, identification of storage fungi, control of seed-borne diseases, seed treatment methods	1
11	Maintenance of aseptic conditions and sterilizationtechniques, Preparation of nutrient stocks for synthetic media	1
12	Preparation of MS medium for micro-propagation and Callus induction	1
13	Preparation of MS medium for micro-propagation and Callus induction	1
14	Inoculation of explants for micro-propagation	1
15	Inoculation of explants for callus induction and subsequently regeneration of	1

	plantlets from matured seeds of field and horticultural crops	
16	Synthetic seed preparation.	1

Suggested readings

1. Agarwal, R.L. 1997. Seed Technology. 2nd edn. Oxford & IBH.
2. McDonald, M.B. Jr and Copeland, L.O. 1997. Seed Production: Principles and Practices.Chapman & Hall
3. Thompson, J.R. 1979. An Introduction to Seed Technology. Leonard Hill.
4. Singhal, N.C. 2003. Hybrid Seed Production in Field Crops. Kalyani.
5. Justice, O.L. and Bass, L.N. 1978. Principles and Practices of Seed Storage. Castle House Publ.Ltd.
6. Tunwar, N.S. and Singh S.N. 1988. Indian Minimum Seed Certification Standards. CSCB,Ministry of Agriculture, New Delhi.
7. Chawla, H.S. 2008. Introduction to Plant Biotechnology. 2nd edn. Oxford &IBH publishingCo. Ltd. 113-B Shahpur Jat, New Delhi-110049.

GPB-212	**Principles of Genetics**	**3(2+1)**

Objective

To make the students acquainted with both principles and practices in the areas of classical genetics, modern genetics, quantitative genetics and cytogenetics.

Theory

Pre and post Mendelian concepts of heredity, Mendelian principles of heredity, Study of model organisms (Drosophila, Arabidopsis, Garden pea, *E. coli*, and mice), Architecture of chromosomes, chromonemata, chromosome matrix, chromomeres, centromere, secondary constriction and telomere, special types of chromosomes, Chromosomal theory of inheritance-cell cycle and cell division-mitosis and meiosis. Probabilit and Chi-square. Types of DNA and RNA, Dominance relationships, Epistatic interactions with example, Introduction and definition of cytology, geneticsand cytogenetics and their interrelation.

Multiple alleles, pleiotropism and pseudoalleles, Sex determination and sex linkage, sex limited and sex influenced traits, Blood group genetics, Linkage and its estimation, crossing over mechanism, chromosome mapping, Structural and numerical variations in chromosomes and their implications, Use of haploids, dihaploids and double haploids in Genetics, Mutation, classification, Methods of inducing mutations, mutagenic agents and induction of mutation. Qualitative and quantitative traits, Polygenes and continuous variations, multiple factor hypothesis, Cytoplasmic inheritance, Nature, structure and replication of genetic material, Protein synthesis, Transcription and translational mechanism of genetic material, Gene concept:

Gene structure, function and regulation.

Practical

Study of microscope, Study of cell structure, Mitosis and Meiosis cell division, Experiments on monohybrid, dihybrid, trihybrid, test cross and back cross, Experiments on epistatic interactions including test cross and back cross, Practice on mitotic and meiotic cell division, Experiments on probability and chi-square test, Determination of linkage and croo-over analysis (through two point test cross data), Study on sex linked inheritance in Drosophila. Study on models on DNA and RNA structures.

Lecture Schedule: Theory

S.N.	Topic	No. of lectures
1	Pre and Post Mendelian concepts of heredity	1
2	Mendelian principles of heredity	1
3	Cell division – mitosis	1
4	Cell division – meiosis	1
5	Probability and Chi-square	1
6	Dominance relationships and gene interaction	1
7	Epistatic gene interactions with examples (complementary, supplementary, duplicate gene interactions)	1
8	Epistatic gene interactions with examples (masking, inhibitory, polymeric and additive gene interactions)	1
9	Pleiotropism, pseudoalleles, Multiple alleles and Blood group genetics	1
10	Sex determination	1
11	Sex limited, sex influenced and sex linked traits	1
12	Sex linkage	1
13	Linkage and its estimation	1
14	Crossing over : introduction & mechanisms	1
15	Chromosome mapping	1
16	Structural changes in chromosome	1
17	Numerical changes in chromosome	1
18	Mutation: introduction, characteristics & classification	1
19	Mutagenic agents: physical and chemical mutagens	1
20	Induction of mutation, Methods of inducing mutation & CIB technique	1
21	Qualitative & Quantitative traits, Polygenes and continuous variations	1
22	Multiple factor hypothesis	1

23	Cytoplasmic inheritance	1
24	Genetic disorders	1
25	Nature, structure and types of genetic material	1
26	Proof for DNA as genetic material	1
27	Replication of genetic material	1
28	Genetic code & Protein synthesis	1
29	Transcription mechanism of genetic material	1
30	Translational mechanism of genetic material	1
31	Gene concept: Gene structure and function	1
32	Gene regulation, operon concept, Lac and Trp operons	1

Lecture Schedule: Practical

S.N.	Topic	No. of lectures
1	Study of microscope: parts and types	1
2	Study of cell structure	1
3	Experiments on monohybrid, test cross and back cross	1
4	Experiments on dihybrid, test cross and back cross	1
5	Experiments on trihybrid, test cross and back cross	1
6	Experiments on epistatic interactions including test cross and back cross	1
7	Experiments on epistatic interactions including test cross and back cross	1
8	Stains and their preparation	1
9	Fixatives and their preparation	1
10	Practice on mitotic cell division	1
11	Practice on meiotic cell division	1
12	Experiments on probability	1
13	Experiments on Chi-square test	1
14	Determination of linkage and cross over analysis (through two point test cross and three point test cross data)	1
15	Study on sex linked inheritance in Drosophila	1
16	Study of models on DNA and RNA structure	1

Suggested readings

1. Fundamentals of Genetics: B. D. Singh
2. Genetics: M. W. Strickberger.
3. Principles of Genetics: Gardner, Simmons and Snustad.
4. Principles of Genetics: Sinnott, Dunn and Dobzhansky

AGEXT-211	Fundamentals of Extension Education	2(1+1)

Objectives

1. State the importance of extension education in agriculture
2. Familiarize with the different types of agriculture and rural development programs launched by govt. of India
3. Classify the types of extension teaching methods
4. Elaborate the importance and different models of communication
5. Explain the process and stages of adoption along with adopters' categories

Theory

Education: Meaning, definition and Types; Extension Education: meaning, definition, scope and process; objectives and principles of Extension Education; Extension Programme planning: Meaning, Process, Principles and Steps in Programme Development. Extension systems in India: extension efforts in pre-independence era (Sriniketan, Marthandam, Firka Development Scheme, Gurgaon Experiment, etc.) and post-independence era (Etawah Pilot Project, Nilokheri Experiment, etc.); Reorganised Extension System (T&V system) various extension/ agriculture development programs launched by ICAR/ Govt. of India (IADP, IAAP, HYVP, KVK, IVLP, ORP, ND, NATP, NAIP, etc.). Social Justice and poverty alleviation programme: ITDA, IRDP/SGSY/NRLM. Women Development Programme: RMK, MSY etc. New trends in agriculture extension: privatization extension, cyber extension/ e-extension, market-led extension, farmer-led extension, expert systems, etc., Attributes of Innovation, DWCRA, Commodity Interest Groups (CIGs)., Farmers Producer Group (FPG). Rural Development: concept, meaning, definition; various rural development programs launched by Govt. of India. Community Development: meaning, definition, concept and principles, Philosophy of C.D. Rural Leadership: concept and definition, types of leaders in rural context; Method of identification of Rural Leader. Extension administration: meaning and concept, principles and functions. Monitoring and evaluation: concept and definition, monitoring and evaluation of extension programs; transfer of technology: concept and models, capacity building of extension personnel; extension teaching methods: meaning, classification, individual, group and mass contact methods, ICT Applications in TOT (New and Social Media), media mix strategies; communication: meaning and definition; Principles and Functions of Communication, models and barriers to communication. Agriculture journalism; diffusion and adoption of innovation: concept and meaning, process and stages of adoption, adopter categories.

Practical

To get acquainted with university extension system. Group discussion- exercise; Identification of rural leaders in village situation; preparation and use of AV aids, preparation of extension literature (leaflet, booklet, folder, pamphlet news stories and success stories); Presentation skills exercise; micro teaching exercise; A visit to village to understand the problems being encountered by the villagers/ farmers; to study organization and functioning of DRDA/PRI and other development departments at district level; visit to NGO/FO/FPO and learning from their experience in rural development; understanding PRA techniques and their application in village development planning; exposure to mass media: visit to community radio and television studio for understanding the process of programme production; script writing, writing for print and electronic media, developing script for radio and television

AGRON-211	**Crop Production Technology–I (*Kharif* crops)**	**3(1+2)**

Objectives

1. To impart basic and fundamental knowledge on principles and practices of kharif crop production
2. To impart knowledge and skill on scientific crop production and management

Theory

Origin, geographical distribution, economic importance, soil and climatic requirements, varieties, cultural practices and yield of Kharif crops. Cereals- rice, maize, sorghum, pearl millet, finger millet and other minor millets, pulses- mothbean, pigeonpea, mungbean and urdbean; oilseeds groundnut, soybean, sesame, castor; fibre crops- cotton and jute; forage crops- sorghum, cowpea, cluster bean, maize, guinea and napier.

Practical

Nursery preparation of different crops, Transplanting of different crop, sowing of soybean, pigeon pea and mungbean, maize, groundnut and cotton, effect of seed size on germination and seedling vigour of *Kharif* crops, effect of sowing depth on germination of *Kharif* crops, identification of weeds in *Kharif* crops, top dressing and foliar feeding of nutrients, study of yield contributing characters and yield calculation of *Kharif* crops, study of crop varieties and important agronomic experiments at experiential farm, recording biometric observations, Study of forage experiments, morphological description of *Kharif* crops, silage and hay making, visit to research centres of related crops.

Lecture schedule: Theory

S.No.	Topic	No. of lectures
1.	Rice-Origin, geographical distribution, economic importance, soil and climatic requirements,	1
2.	Rice- varieties, cultural practices and yield	1

3.	Maize-Origin, geographical distribution, economic importance, soil and climatic requirements, varieties, cultural practices and yield	1
4.	Sorghum-Origin, geographical distribution, economic importance, soil and climatic requirements,	1
5.	Sorghum- varieties, cultural practices and yield (Seed and forage)	1
6.	Pearl millet- Origin, geographical distribution, economic importance, soil and climatic requirements, varieties, cultural practices and yield	1
7.	Finger millet- Origin, geographical distribution, economic importance, soil and climatic requirements, varieties, cultural practices and yield	1
8.	Minor millet: Origin, geographical distribution, economic importance, soil and climatic requirements, varieties, cultural practices and yield	1
9.	Mothbean, Mungbean and Urdbean -Origin, geographical distribution, economic importance, soil and climatic requirements, varieties, cultural practices and yield	1
10.	Pigeonpea-Origin, geographical distribution, economic importance, soil and climatic requirements, varieties, cultural practices and yield	1
11.	Groundnut-Origin, geographical distribution, economic importance, soil and climatic requirements, varieties, cultural practices and yield	1
12.	Soybean-Origin, geographical distribution, economic importance, soil and climatic requirements, varieties, cultural practices and yield	1
13.	Sesame and Castor -Origin, geographical distribution, economic importance, soil and climatic requirements, varieties, cultural practices and yield	1
14.	Cotton-Origin, geographical distribution, economic importance, soil and climatic requirements, varieties, cultural practices and yield	1
15.	Jute-Origin, geographical distribution, economic importance, soil and climatic requirements, varieties, cultural practices and yield	1
16.	Forage crops- sorghum, cowpea, cluster bean, maize, guinea and napier, package of practices	1

Lecturer schedule: Practical

S.No.	Topic	No. of lectures
1.	Nursery preparation of different crop	3
2.	Transplanting of different crop	3
3.	Sowing of soybean, pigeon pea and mungbean, maize, groundnut and cotton	2

4.	Sowing of maize, groundnut and cotton	2
5.	Effect of seed size on germination and seedling vigour of *Kharif* crops	2
6.	Effect of sowing depth on germination of *Kharif* crops	2
7.	Working out seed rate and real value related numericals	2
8.	Identification of weeds in *Kharif* crops	2
9.	Top dressing and foliar feeding of nutrients	2
10.	Study of yield contributing characters and yield calculation of *Kharif* crops	2
11.	Study of crop varieties and important agronomic experiments at experiential farm	2
12.	Recording biometric observations	2
13.	Study of forage experiments	2
14.	Morphological description of *Kharif* crops	2
15.	Silage and hay making, visit to research centres of related crops	2

Suggested Readings

1. B. Gurarajan, R. Balasubramanian and V. Swaminathan. Recent Strategies on Crop Production. Kalyani Publishers, New Delhi.
2. Chidda Singh.1997. Modern techniques of raising field crops. Oxford and IBH Publishing Co. Pvt. Ltd., New Delhi.
3. Rajendra Prasad. Textbook of Field Crops Production - Commercial Crops. Volume II ICAR Publication.
4. S.R. Reddy. 2009. Agronomy of Field Crops. Kalyani Publishers, New Delhi.
5. S.S. Singh. 2005. Crop Management. Kalyani Publishers, New Delhi.
6. UAS, Bangalore. 2011. Package of Practice. UAS, Bangalore.
7. Subhash Chandra Bose, M. and Balakrishnan, V. 2001. Forage Production. South Asian Publishers, New Delhi.

AGRON-212	**Principles and Practices of Natural Farming**	**2(1+1)**

Objectives

1. To provide comprehensive understanding and knowledge to students about natural farming.
2. To teach students the concept, need and principles of native ecology-based production under natural farming.
3. To impart practical knowledge of natural farming and related agricultural practices in Indian and global environmental and economic perspectives.

Theory

Indian Heritage of Ancient Agriculture, History of Natural Farming, Importance of natural farming in view of climate change, soil health, water use carbon sequestration, biodiversity conservation, food security and nutritional security, and sustainable development goals (SDGs), Concept of natural farming; Definition of natural farming; Objective of natural farming, Essential characteristics and Principles of natural farming; Scope and importance of natural farming. Main Pillars of natural farming; Methods/ types/schools of natural farming. Characteristics and design of a natural farm, Concept of ecological balance, ecological engineering and community responsibility in natural versus other farming systems, Introduction to concept of ecological, water, carbon and nitrogen foot prints, Concept and evaluation of ecosystem services, integration of crops, trees and animals, cropping system approaches, Biodiversity, indigenous seed production, farm waste recycling, water conservation and renewable energy use approaches on a natural farm, Rearing practices for animals under natural farming, Nutrient management in natural farming and their sources, Insect, pest, disease and weed management under natural farming; Mechanization in natural farming, Processing, labelling, economic considerations and viability, certification and standards in natural farming, marketing and export potential of natural farming produce and products. Initiatives taken by Government (central/state), NGOs and other organizations for promotion of natural farming and chemical free agriculture, Case studies and success stories in natural farming and chemical free traditional farming, Entrepreneurship opportunities in natural farming.

Practical

Visit of natural farm and chemical free traditional farms to study the various components and operations of natural farming principles at the farm; Indigenous technical knowledge (ITK) for seed, tillage, water, nutrient, insect-pest, disease and weed management; On-farm inputs preparation methods and protocols, Studies in green manuring in-situ and green leaf manuring, Studies on different types of botanicals and animal urine and dung based non-aerated and aerated inputs for plant growth, nutrient, insect and pest and disease management; Weed management practices in natural farming; Techniques of Indigenous seed production- storage and marketing, Partial and complete nutrient and financial budgeting in natural farming; farming; Evaluation of ecosystem services in natural farming (Crop, Field and System).

Lecture schedule: Theory

S.N.	Topic	No. of lectures
1.	Indian Heritage of Ancient Agriculture, History of Natural Farming, Importance of natural farming in view of climate change, soil health, water use carbon sequestration, biodiversity conservation, food security and nutritional security, and sustainable development goals (SDGs)	2
2.	Concept of natural farming; Definition of natural farming; Objective of	1

	natural farming, Essential characteristics and Principles of natural farming; Scope and importance of natural farming	
3.	Main Pillars of natural farming; Methods/ types/schools of natural farming. Characteristics and design of a natural farm	1
4.	Concept of ecological balance, ecological engineering and community responsibility in natural versus other farming systems	1
5.	Introduction to concept of ecological, water, carbon and nitrogen foot prints	1
6.	Concept and evaluation of ecosystem services, integration of crops, trees and animals, cropping system approaches	1
7.	Biodiversity, indigenous seed production, farm waste recycling, water conservation and renewable energy use approaches on a natural farm	1
8.	Rearing practices for animals under natural farming	1
9.	Nutrient management in natural farming and their sources	1
10	Insect, pest, disease and weed management under natural farming	1
11.	Mechanization in natural farming, Processing, labelling, economic considerations and viability	1
12.	Certification and standards in natural farming	1
13	marketing and export potential of natural farming produce and products	1
14	Initiatives taken by Government (central/state), NGOs and other organizations for promotion of natural farming and chemical free agriculture	1
15	Case studies and success stories in natural farming and chemical free traditional farming, Entrepreneurship opportunities in natural farming	1

Lecture schedule: Practical

S.N.	Topic	No. of lectures
1.	Visit of natural farm and chemical free traditional farms to study the various components and operations of natural farming principles at the farm	2
2.	Indigenous technical knowledge (ITK) for seed, tillage, water, nutrient, insect-pest, disease and weed management	3
3.	On-farm inputs preparation methods and protocols	2
4.	Studies in green manuring in-situ and green leaf manuring	2
5.	Studies on different types of botanicals and animal urine and dung based non-aerated and aerated inputs for plant growth, nutrient, insect and pest and disease management	2

6.	Weed management practices in natural farming	1
7.	Techniques of Indigenous seed production- storage and marketing	1
8.	Partial and complete nutrient and financial budgeting in natural farming	1
9.	Evaluation of ecosystem services in natural farming (Crop, Field and System)	2

Suggested readings

1. Ayachit, S.M. 2002. Kashyapi Krishi Sukti (A Treatise on Agriculture by Kashyapa). Brig Sayeed Road, Secunderabad, Telangana: Asian Agri-History Foundation 4: 205.
2. Boeringa, R. (Eed.). 1980. Alternative Methods of Agriculture. Elsevier, Amsterdam, 199 pp.
3. Das, P., Das, S.K., Arya, H.P.S., Reddy, G. Subba, Mishra, A. and others: Inventory of Indigenous Technical Knowledge in Agriculture: Mission mode Project on Collection, Documentation and Validation of Indigenous Technical Knowledge, Document 1 To 7, Indian Council of Agricultural Research, New Delhi.
4. Ecological Farming -The seven principles of a food system that has people at its heart. May 2015, Greenpeace.
5. Ecological Farming, The Seven principles of a food system that has people at its heart. May 2015, Greenpeace
6. FAO. 2018. The 10 elements of agro-ecology: guiding the transition to sustainable food and agricultural system.https://www.fao.org/3/i9037en/i9037en.pdf Agro ecosystem Analysis for Research and Development Gordon R. Conway.1985.
7. Fukuoka, M. 1978. The One-Straw Revolution: An Introduction to Natural Farming. Rodale Press, Emmaus, PA. 181 pp
8. Fukuoka, M. 1985. The Natural Way of Farming: The Theory and Practice of Green Philosophy. Japan Publications, Tokyo, 280 pp.
9. Hill S.B and Ott. P. (Eeds.). 1982. Basic Techniques in Ecological Farming Berkhauser Verlag, Basel, Germany, 366 pp.
10. Hill, S.B. and Ott, P. (Eeds.). 1982. Basic Techniques in Ecological Farming. Berkhauser Verlag, Basel, Germany, 366 pp.
11. HLPE. 2019. Agroeeological and other innovative approaches for sustainable agriculture and food systems that enhance food security and nutrition. A report by the High Level Panel of Experts on Food Security and nutrition of the Committee on World Food Security, Rome. https://fao.org/3/ea5602en/ea5602en.pdf.
12. INFRC. 1988. Guidelines for Nature Farming Techniques. Atami, Japan. 38 pp.
13. Khurana, A. and Kumar, V. 2020. State of Organic and Natural Farming: Challenges and Possibilities, Centre for Science and Environment, New Delhi.
14. Malhotra R. and S.D. Babaji. 2020. Sanskrit Non Translatable- The importance of Sanskritizing English. Amaryllis, New Delhi India.
15. Nalini, S. 1996. Vrikshayurveda (The Science of Plant Life) by Surapala. AAHF Classic Bulletin 1. Asian Agri-History Foundation, Brig Sayeed Road, Secunderabad, AP (now Telengana), India. 94pp.
16. Nalini, S. 1999. Krishi-Parashara (Agriculture by Parashara) by Parashara. Brig Sayeed Road, Secunderabad, Telangana: AAHF Classic Bulletin, Asian Agri-History Foundation. 104pp.
17. Nalini, S. 2011. Upavana Vinoda (Woodland Garden for Enjoyment) by Sarangdhara (13th century CE): AAHF Classic Bulletin 8. Asian Agri-History Foundation, Brig Sayeed Road,

Secunderabad, AP (now Telangana), India. 64p

18. Natural Asset Farming: Creating Productive and Biodiverse Farms by David B. Lindenmayer, Suzannah M. Macbeth, et al. (2022)
19. Natural Farming Techniques: Farming without tilling by Prathapan Paramu (2021)
20. Plenty for All: Natural Farming A to Z Prayog Pariwar Methodology by Prof. Shripad A. Dabholkar and Prayog Pariwar Prayog Pariwar (2021)
21. Reyes Tirado. 2015. Ecological Farming- The seven principles of a food system that has people at its heart. Greenpeace Research laboratories. University of Exeter, Ottho Heldringstraat.
22. Shamasastry, R. 1915. Kautilya's Arthashastra.
23. The Ultimate Guide to Natural Farming and Sustainable Living: Permaculture for Beginners (Ultimate Guides) by Nicole Faires (2016)
24. U. K. Behera. 2013. A text Book of Farming System. Agrotech Publishing House, Udaipur.
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AGINFO-211	Agricultural Informatics and Artificial Intelligence	3(2+1)

Objective

i) To acquaint student with the basics of computer applications in agriculture, multimedia, data base management, application of mobile app and decision- making processes, etc.

ii) To provide basic knowledge of computer with applications in Agriculture.

iii) To make the students familiar with Agricultural-Informatics, its components and applications in agriculture.

Theory

Introduction to Computers, Anatomy of Computers, Memory Concepts, Units of Memory, Operating System: Definition and types, Applications of MS-Office forcreating, Editing and Formatting a document, Computer programming: General concepts, Introduction to Visual Basic, Java, Fortran, C/C++, etc. concepts and standard input/ output operations.

e-Agriculture, Concepts, design and development, Application of innovative ways to use information and communication technologies (IT) in Agriculture, Computer Models in Agriculture: Statistical,weather analysis and crop simulation models, concepts, structure, inputs-outputs files, limitation, advantages and applicationof models for understandingplant processes,sensitivity, verification, calibrationandvalidation, IT applications for computation of water and nutrient requirement of crops, e-Agriculture, Concepts, design and development, Application of innovative ways to use information and communication technologies (IT) in Agriculture, Computer Models in Agriculture: Statistical,weather analysis and crop simulation models, concepts, structure, inputs-outputs files, limitation, advantages and applicationof models for understandingplant processes,sensitivity, verification, calibrationandvalidation, IT applications for computation of water and nutrient requirement of crops etc.

S.No.	Topic	No. of Lecture
1	Introduction to Computers, Anatomy of Computers, Memory Concepts, Units of Memory, Operating System: Definition and types, Applications of MS-Office for creating, Editing and Formatting a document	5
2	Computer programming: General concepts, Introduction to Visual Basic, Java, Fortran, C/C++, etc. concepts and standard input/ output operations.	5
3	e-Agriculture, Concepts, design and development, Application of innovative ways to use information and communication technologies (IT) in Agriculture,	5
4	Computer Models in Agriculture: Statistical, weather analysis and crop simulation models, concepts, structure, inputs-outputs files, limitation, advantages and application of models for under standing plant processes, sensitivity, verification, calibration and validation,	6
5	Computer Models in Agriculture: Statistical,weather analysis and crop simulation models, concepts, structure, inputs-outputs files, limitation, advantages and application of models for under standing plant processes, sensitivity, verification, calibration and validation,	6
6	IT applications for computation of water and nutrient requirement of crops, e-Agriculture, Concepts, design and development, Application of innovative ways to use information and communication technologies (IT) in Agriculture, Computer Models in Agriculture: Statistical,weather analysis and crop simulation models, concepts, structure, inputs-outputs files, limitation, advantages and application of models for under standing plant processes, sensitivity, verification, calibration and validation	6
7	IT applications for computation of water and nutrient requirement of Crops etc.	5

HORT-211	Production Technology of Fruit and Plantation Crops	2(1+1)

Objectives

1. To educate about the different forms of classification of fruit crops 2 (1+1)
2. To educate about the origin, area, climate, soil, improved varieties and cultivation practices of fruit and plantation crops
3. To educate about the physiological disorders of fruit crops, palms and plantation crops

Theory

Production status of fruit and plantation crops: Importance and scope of fruit and plantation crop industry in India; nutritional value of fruit crops; classification of fruit crops; area, production, productivity and export potential of fruit and plantation crops. Crop production techniques in tropical, sub-tropical and temperate fruit crops: Climate and soil requirements, varieties, propagation and use of rootstocks, planting density and systems of planting: High density and ultra-high-density planting, cropping systems, after care – training and pruning; water, nutrient and weed management, fertigation, special horticultural techniques, plant growth regulation, important disorders, maturity indices and harvest, value addition.

Fruit crops: mango, banana, papaya, guava, sapota, citrus, grape, litchi, pineapple, pomegranate, apple, pear, peach, strawberry, nut crops Jackfruit and minor fruits- date palm, ber, aonla, plantation crops-coconut, arecanut, cashew, tea, coffee and rubber.

Crop production techniques in palms and plantation crops: Climate and soil requirements, varieties, propagation, nursery management, planting and planting systems, cropping systems, after care, training and pruning for plantation crops, water, nutrient and weed management, intercropping, multi-tier cropping system, mulching, special horticultural practices, maturity indices, harvest and yield, pests and diseases, processing- value addition.

Palms: coconut, arecanut, oil palm and palmyrah, plantation crops: tea, coffee, cocoa, cashewnut, rubber.

Practical

Propagation techniques, selection of planting material, varieties, important cultural practices for mango, banana, papaya, guava, sapota, grapes, Citrus (mandarin and acid lime), pomegranate, jackfruit, preparation and application of PGR's for propagation, Micro propagation, protocol for mass multiplication and hardening of fruit crops, Identification and description of varieties, mother palm and seed nut selection, nursery practices, seedling selection, fertilizers application, nutritional disorders, pests and diseases of Coconut, Arecanut and cocoa, Tea and coffee, Rubber and cashew, Visit to commercial orchard and plantation industries.

Lecture schedule: Theory

S. No.	Topics	No. of lectures
1.	Production status of fruit and plantation crops (area, production,	1

	productivity and export potential of fruit and plantation crops)	
2.	Importance and scope of fruit and plantation crop industry in India; Nutritional value of fruit crops	1
3.	Classification of fruit crops	1
4.	Crop production techniques in tropical fruit crops - Mango	1
5.	Banana	1
6.	Papaya	1
7.	Guava	1
8.	Sapota and Jackfruit	1
9.	Sub-tropical fruit crops - Citrus	1
10.	Grape	1
11.	Litchi, Pomegranate, Pineapple	1
12.	Temperate fruit crops-Apple	1
13.	Pear, Peach, Strawberry	1
14.	Minor fruits- Date palm, Ber, Aonla, Bael	1
15.	Palms and plantation crops - Coconut, Arecanut, Oil Palm and Palmyrah	1
16.	Plantation crops - Tea, Coffee, Cocoa, Cashewnut, Rubber	1
		16

Lecture schedule: Practical

S. No.	**Topics**	**No. of lectures**
1.	Propagation techniques of fruit and plantation crop	2
2.	Selection of planting material, varieties, important cultural practices - mango	1
3.	Papaya	1
4.	Banana	1
5.	Guava	1
6.	Sapota and grapes	1
7.	Citrus (mandarin and acid lime), pomegranate and jackfruit	1
8.	Preparation and application of PGR’s for propagation.	1
9.	Micro propagation, mass multiplication and hardening of fruit crops	1
10.	Identification and description of varieties, mother palm and seed nut selection	1

11.	Nursery practices, seedling selection, fertilizers application	2
12.	Nutritional disorders of fruit and plantation crop	1
13.	Pests and diseases of fruit and plantation crop	1
14.	Visit to commercial orchard/plantation industries	1
		16

Suggested Readings

1. Banday, F.A. and Sharma, M.K. 2010 Advances in temperate fruit production. Kalyani Publishers, Ludhiana.
2. Bose, T.K., S.K. Mitra and D. Sanyal 2001. Fruits: Tropical and Subtropical (2 volumes) Naya Udyog, Calcutta.
3. Bose, T.K., S.K. Mitra, A.A. Farooqi and M.K. Sadhu (Eds). 1999. Tropical Horticulture Vol.1. Naya Prokash, Calcutta.
4. Chadha, K.L. 2001. Handbook of Horticulture. ICAR, Delhi.
5. Chadha, T.R. 2001 Textbook of temperate fruits. ICAR, New Delhi
6. Chattopadhyay, T.K. 2001. A Text Book on Pomology (4 volumes). Kalyani Publishers, Ludhiana.
7. Chattopadhyay. 1998. A textbook on pomology (sub-tropical fruits) vol. III. Published by M/s. Kalyani publishers, Ludhiana, New Delhi, Noida. UP.
8. Chudawat, B. S.1990. Arid fruit culture Oxford &IBH, New Delhi
9. David Jackson and N.E. Laone, 1999. Subtropical and temperate fruit production. CABI publications.
10. Kumar, N. 1997. Introduction to Horticulture. Rajalakshmi Publications, Nagercoil, Tamil Nadu.
11. Mitra, S.K., T.K. Bose and D.S. Rathore. 1991. Temperate fruits. Horticulture and allied Publishers, Calcutta.
12. Pal, J.S. 1997. Fruit Growing. Kalyani Publishers, New Delhi.
13. Radha, T. and Mathew, L.2007. Fruit crops. New India publishing Agency.
14. Sadhu, M.K. and P.K. Chattopadhyay. 2001. Introductory Fruit Crops. Naya Prokash, Calcutta.
15. Singh, S.P. 2004. Commercial Fruits. Kalyani Publishers, Ludhiana.
16. Veeraragavathatham, D., Jawaharlal, M., Jeeva, S., Rabindran, R and Umapathy, G. 2004 (2nd edition). Scientific fruit culture. Published by M/s. Suri associates, 1362/4, Velraj Vihar Complex, Thadagam Road, Coimbatore- 2
17. W.S. Dhillon. 2013. Fruit production in India. Narendra publishing House, New Delhi
18. Kavino, M, V. Jegadeeswari, R. M. Vijayakumar and S. Balkrishnan. 2018. Production Technology of Fruits and Plantation Crops by Narendra Publishing House.
19. Kumar, N.J. B.M. Md. Abdul Khaddar, Ranga Swamy, P. and Irulappan, I. 1997. Introduction to spices, Plantation crops and Aromatic plants. Oxford & IBH, New Delhi.
20. Sharma, A., Kumar, P., Tripathi, V.K. 2024. Production Technology of Fruits and Plantation Crops. Elite Publishing House
21. Thampan, P.K. 1981. Handbook of coconut palm. Oxford & IBH, New Delhi.
22. Thompson, P.K. 1980. Coconut. Oxford & IBH, New Delhi
23. V. Ponnuswami, M. Kumar; S. Ramesh Kumar and C. Krishnamoorthy, 2015. Fruit and Plantation Crops Narendra Publishing House.
24. Yadav, P. K. 2007. Fruit Production Technology, Published by International Book Distributing Company, Lucknow ISBN 81-8189-206-2
25. Yadav, P. K. and Singh, J. 2000. Phalotpadan and Prasanskaran Published.in 2000 from Agro- bios (India), Jodhpur., ISBN 81-7754-086-6
26. Yadav, P.K. 2021. Production Technology of Tropical and Sub Tropical Fruit Crops, New India Publishing Agency ,New Delhi, ISBN 978-93-90512-29-4

27. Yadav, P.K. 2021. Production Technology of Fruit and Plantation Crops, New India Publishing Agency ,New Delhi, ISBN 978-93-90512-29-4

PPATH-211	Fundamentals of Nematology	2(1+1)

Objectives

1. To impart knowledge on history, economic importance of plant parasitic nematodes, morphology, biology, host parasitic relationship of nematodes.
2. To impart knowledge on nematode pests of different crops of national and local importance and their management.

Theory

Introduction: History of phytonematology, habitat and diversity, economic importance of nematodes. General characteristics of plant parasitic nematodes. Nematode: definition, general morphology and biology. Classification of nematodes up to family level with emphasis on groups containing economically important genera. Classification of nematodes on the basis of feeding/ parasitic habit. Symptomatology, role of nematodes in disease development, Interaction between plant parasitic nematodes and disease-causing fungi, bacteria and viruses. Nematode pests of crops: Rice, wheat, vegetables, pulses, oilseed and fiber crops, citrus and banana, tea, coffee and coconut. Different methods of nematode management: Cultural methods, physical; methods, Biological methods, Chemical methods, Plant Quarantine, Plant resistance and INM.

Practical

Sampling methods, collection of soil and plant samples; Extraction of nematodes from soil and plant tissues following Cobb's sieving and decanting technique, Baermann funnel technique, Picking and counting of plant parasitic nematode. Identification of economically important plant nematodes up to generic level with the help of keys and description: Meloidogyne, Pratylenchus; Heterodera, Tylenchulus, Xiphinema, and Helicotylenchus etc. Study of symptoms caused by important nematode pests of cereals, vegetables, pulses, plantation crops etc. Methods of application of nematicides and organic amendments.

Lecture Schedule: Theory

S.No.	Topic	No. of lectures
1.	Introduction: History of phytonematology, habitat and diversity, economic importance of nematodes.	01
2.	General characteristics of plant parasitic nematodes. Nematode: definition, general morphology and biology.	01
3.	Classification of nematodes up to family level with emphasis on groups	02

	containing economically important genera.	
4.	Classification of nematodes on the basis of feeding/ parasitic habit. Symptomatology, role of nematodes in disease development.	02
5.	Interaction between plant parasitic nematodes and disease-causing fungi, bacteria and viruses.	02
6.	Nematode pests of crops: Rice, wheat, vegetables, pulses, oilseed and fiber crops, citrus and banana, tea, coffee and coconut.	03
7.	Different methods of nematode management: Cultural methods, physical; methods, Biological methods, Chemical methods.	03
8.	Plant Quarantine, Plant resistance and INM.	02

Lecture Schedule: Practical

S.No	**Topic**	**No. of lectures**
1.	Sampling methods, collection of soil and plant samples	02
2.	Extraction of nematodes from soil and plant tissues following Cobb's sieving	01
3.	Extraction of nematodes from soil and plant tissues following decanting technique and Baermann funnel technique	02
4.	Picking and counting of plant parasitic nematode.	02
5.	Identification of economically important plant nematodes up to generic level with the help of keys and description: Meloidogyne, Pratylenchus; Heterodera, Tylenchulus, Xiphinema and Helicotylenchus etc.	03
6.	Study of symptoms caused by important nematode pests of cereals, vegetables, pulses, plantation crops etc.	03
7.	Methods of application of nematicides and organic amendments.	03

Suggested readings

1. Economic Nematology-Edited by J.M. Webster
2. Plant Parasitic Nematodes (Vol-1) by Zukerman, Mai, Rohde
3. Plant Parasitic Nematodes of India: Problems and Progress by - Gopal Swarup, D. R. Dasgupta, P. K. Koshy.
4. Text book on Introductory Plant Nematology -R.K. Walia and H.K. Bajaj.

PHED-211	Physical Education, First Aid, Yoga Practices and Meditation	2(0+2)

Objectives

i) To make the students a ware about Physical Education, First Aid and Yoga Practices

ii) To disseminate the knowledge and skill how to perform physical training, perform first aid and increase stamina and general well being through yoga.

Practical

1. **Physical education;** Training and Coaching – Meaning &Concept; Methods of Training; aerobic and aerobic exercises; Calisthenics, weight training, circuit training, interval training, Fartlek training; Effects of Exercise on Muscular, Respiratory, Circulatory & Digestive systems; Balanced Diet and Nutrition: Effects of Diet on Performance; Physiological changes due to ageing and role of regular exercise on ageing process; Personality, its dimensions and types; Role of sports in personality development; Motivation and Achievements in Sports; Learning and Theories of learning; Adolescent Problems & its Management; Posture; Postural Deformities; Exercises for good posture.
2. **Yoga;** History of Yog, Types of Yog, Introduction to Yog,
   - Asanas (Definition and Importance) Padmasan, Gaumukhasan, Bhadrasan, Vajrajasan, Shashankasan, Pashchimotasan, Ushtrasan, Tadasan, Padhas tasan, Ardhchandrasan, Bhujangasan, Utanpadasan, Sarvangasan , Parvatasan , Patangasan, Shishupalanasan –left leg-right leg, Pavanmuktasan, Halasan, Sarpasan, Ardhdhanurasan, Sawasan.
   - Suryanamskar Pranayama (Definition and Importance) Omkar, Suryabhedan, Chandrabhedan, AnulomVilom, Shitali, Shitkari, Bhastrika, Bhramari.
   - Meditation (Definition and Importance), Yogic Kriyas (Kapalbhati), Tratak, Jalneti and Tribandh.
   - Mudras (Definition and Importance) Gyanmudra, Dhyanmudra, Vayumudra, Akashmudra, Pruthvimudra, Shunyamudra, Suryamudra, Varunmudra, Pranmudra, Apanmudra, Vyanmudra, Uddanmudra.

   - Role of yoga in sports :-Teaching of Asanas–demonstration, practice, correction and practice.
3. **Sports;-** History of sports and ancient games, Governance of sports in India; Important national sporting events; Awards in Sports; History, latest rules,

measurements of play field, specifications of equipment, skill, technique, style and coaching of major games (Cricket, football, table Tennis, Badminton, Volleyball, Basketball, Kabaddi and Kho-Kho) and Athletics. Need and requirement of first aid. First Aid equipment and up keep. First AID Techniques, First aid related with respiratory system. First aid related with Heart, Blood and Circulation. First aid related with Wound sand Injuries. First aid related with Bones, Joints Muscle related injuries. First aid related with Nervous system and Unconsciousness. First aid related with Gastrointestinal Tract. First aid related with Skin, Burns. First aid related with Poisoning. First aid related with Bites and Stings. First aid related with Sense organs, Handling and transport of injured traumatized persons. Sports injuries and their treatments.

NCC-III/NSS- III/ PHED-III	**NSS/NCC/Physical Education &Yoga Practices**	**1(0+1)**

NCC-III	**National Cadet Corps**	**1(0+1)**

1. Arms Drill- Attention, stand at ease, stand easy. Getting on parade. Dismissing and falling out. Ground/take up arms, examine arms.
2. Shoulder from the order and vice-versa, present from the order and vice-versa.
3. Saluting at the shoulder at the halt and on the march. Short/long trail from the order and vice-versa.
4. Guard mounting, guard of honour, Platoon/CoyDrill.
5. Characteristics of rifle (.22/.303/SLR), ammunition, firepower, stripping, assembling, care, cleaning and sight setting.
6. Loading, cocking and unloading. The lying position and holding.
7. Trigger control and firing a shot. Range Procedure and safety precautions. Aiming and alteration of sight.
8. Theory of groups and snap shooting. Firing at moving targets. Miniature range firing.
9. Characteristics of Carbine and LMG.
10. Introduction to map, scales and conventional signs. Topographical forms and technical terms.

NSS- III	**National Service Scheme**	**1(0+1)**

1. Vocational skill development

To enhance the employment potential and to set up small business enterprises skills of volunteers, a list of 12 to 15 vocational skills will be drawn up based on the local conditions and opportunities. Each volunteer will have the option to select two skill-areas out of this list Issues related environment

2. **Environmental conservation**, enrichment and sustainability, climatic change, natural resource management (rain water harvesting, energy conservation, forestation, waste land development and soil conservations) and waste management

3. **Disaster management**

   Introduction and classification of disaster, rehabilitation and management after disaster; role of NSS volunteers in disaster management.

4. **Entrepreneurship development**

   Definition, meaning and quality of entrepreneur; steps in opening of an enterprise and role of financial and support service institution.

5. **Formulation of production oriented project**

   Planning, implementation, management and impact assessment of project.

6. **Documentation and data reporting**

   Collection and analysis of data, documentation and dissemination of project reports.

PHED-III	**Physical Education &Yoga Practices**	**1(0+1)**

1. Teaching of skills of Hockey– demonstration practice of the skills and correction.
2. Teaching of skills of Hockey–demonstration practice of the skills and correction. And involvement of skills in games situation with teaching of rules of the game.
3. Teaching of skills of Kho-Kho– demonstration practice of the skills and correction.
4. Teaching of skills of Kho-Kho–demonstration practice of the skills and correction. involvement of the skills in games situation with teaching of rules of the game
5. Teaching of different track events– demonstration practice of the kills and correction.
6. Teaching of different track events – demonstration practice of the skills and correction with competition among them.

Teaching of different field events–demonstration practice of the skills and correction.